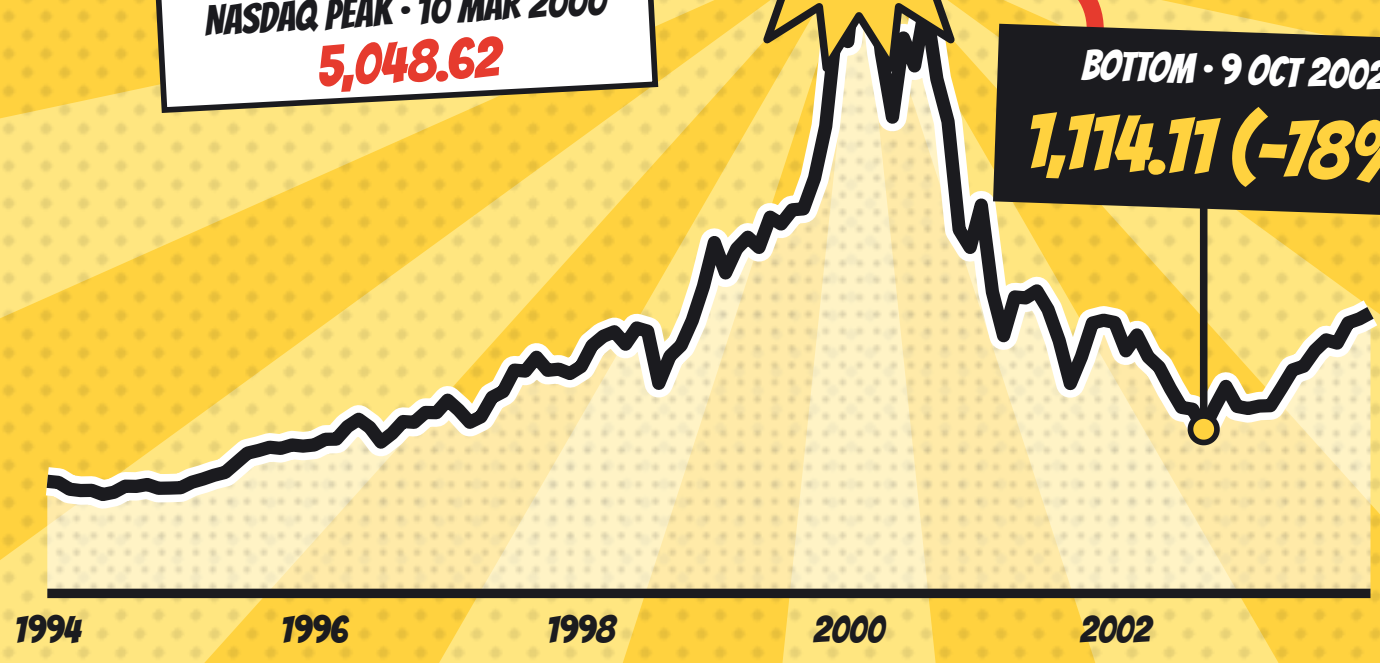
THE DOT-COM BUBBLE!

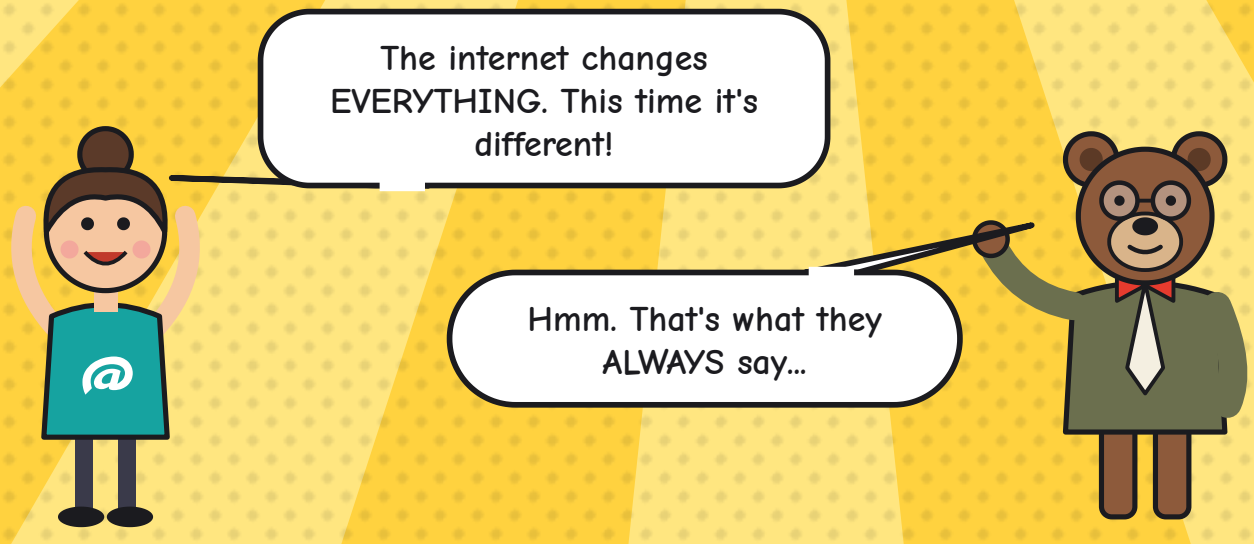
1995 - 2002

POP!

NASDAQ PEAK · 10 MAR 2000
5,048.62

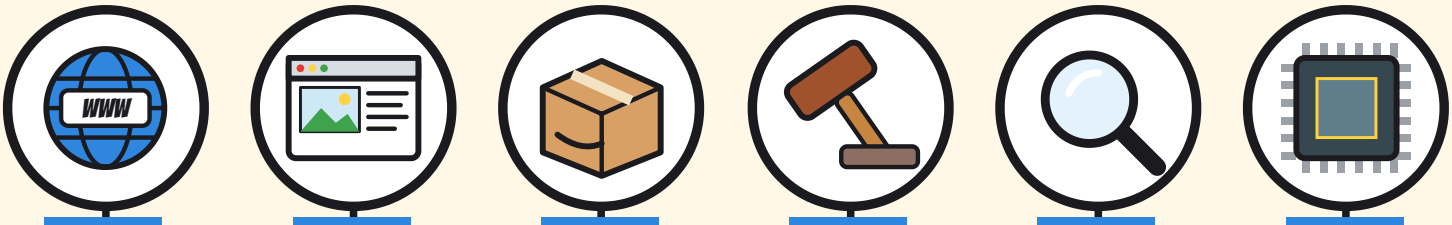
BOTTOM · 9 OCT 2002
1,114.11 (-78%)





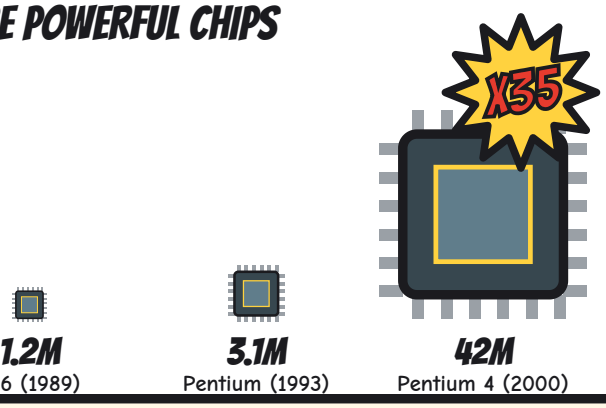
THE TECH REVOLUTION

Where the good times came from: computers got cheap and powerful, and the internet linked them all together.

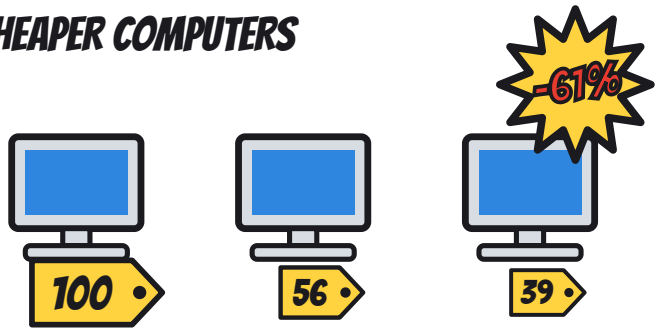


1989 Web proposed at CERN · 1993 CERN makes it free, Mosaic browser · 1994 Amazon · 1995 eBay · 1998 Google · 2000 Intel Pentium 4

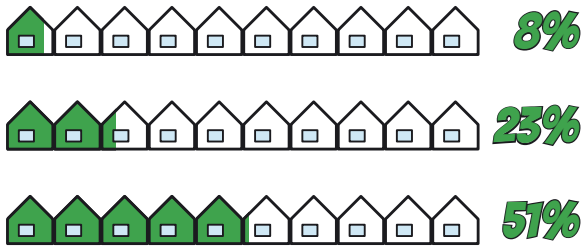
MORE POWERFUL CHIPS



CHEAPER COMPUTERS

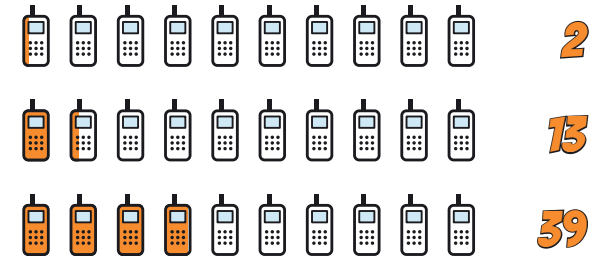


HOMES WITH A COMPUTER

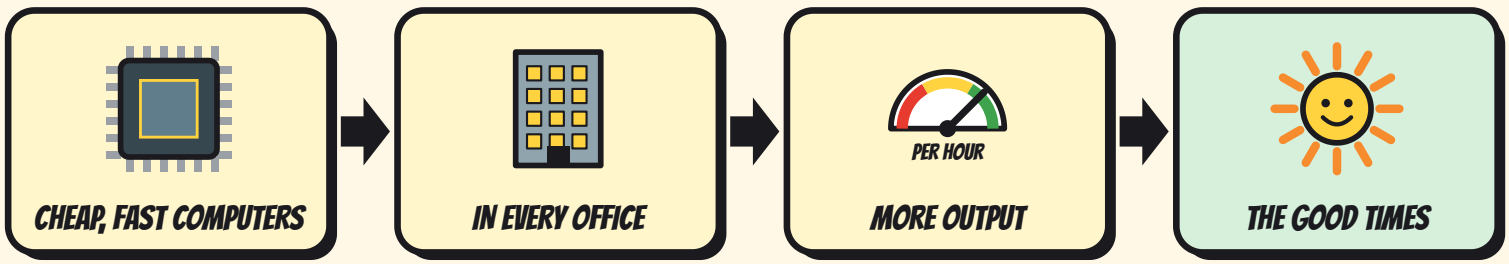


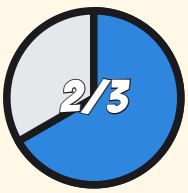
each house = 10% of homes

MOBILE PHONES

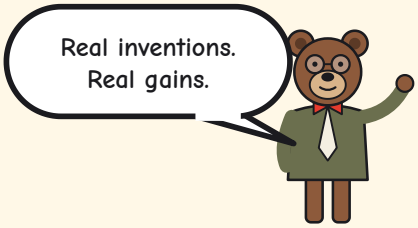


each phone = 10 per 100 people





of the late-1990s productivity speed-up came from making and using computers (Federal Reserve economists)

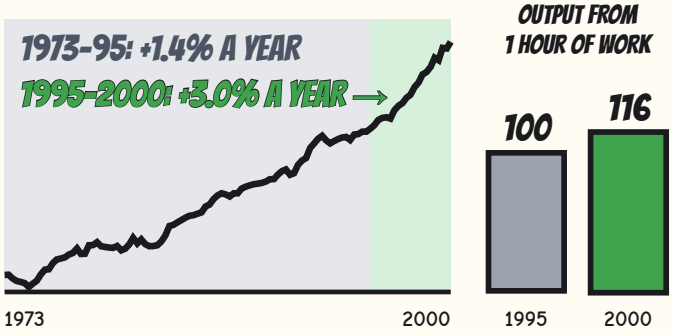




THE GOOD TIMES

Before the bubble, the US economy really was booming. These true facts made people believe the boom could go on forever.

WORKERS GOT MORE PRODUCTIVE



Output per hour, US firms (1995 = 100). Same hour: +16% by 2000.



3.8%

JOBS FOR ALMOST EVERYONE

Unemployment fell to 3.8% in April 2000, the lowest since 1969.



~2%

PRICES STAYED CALM

Consumer prices rose only about 2% a year in 1997–99, so the Fed had no reason to panic.



$236BN

THE GOVERNMENT SAVED MONEY

The US budget was in surplus in 1998–2001, the first since 1969. The 2000 surplus was $236 billion.



9% → 43%

EVERYONE WENT ONLINE

The share of Americans using the internet jumped from 9% in 1995 to 43% in 2000.



1996

NEW RULES OPENED TELECOMS

The Telecommunications Act (8 Feb 1996) let new firms compete in phone and cable markets.



It's a NEW ECONOMY! Shares can only go up!

The good news was REAL. That's what made the bubble so convincing.



FROM BOOM TO BUBBLE

Real progress turned into wild hope. Step by step, share prices raced far ahead of the profits behind them.

9 AUG 1995

NETSCAPE GOES PUBLIC

Priced at $28, its shares closed the first day at $58.25. A 16-month-old company was worth $2.9 billion overnight.

1

5 DEC 1996

"IRRATIONAL EXUBERANCE"

Fed chair Alan Greenspan asked how we would know when prices had gone too far. Markets dipped, then kept climbing for 3 more years.

2

1999

IPO MANIA

On average, new shares jumped about 71% on their first trading day in 1999. The Nasdaq rose 86% in that one year.

3

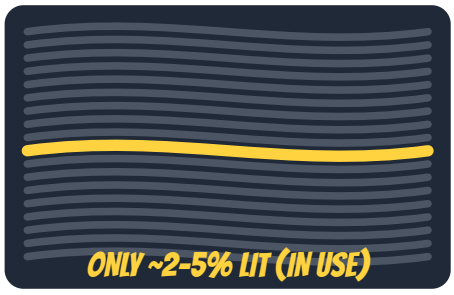
"EYEBALLS"

CLICKS, NOT PROFITS

Investors valued websites by visitors, not earnings. In March 2000, 74% of 207 internet firms studied were burning cash.

4

FIBRE

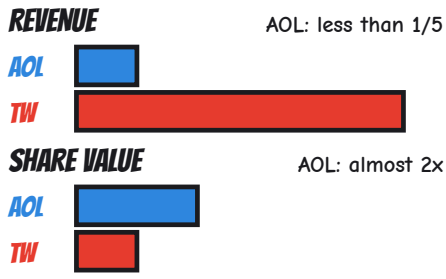


CABLES NOBODY NEEDED (YET)

Telecom firms expected internet traffic to keep exploding. They borrowed heavily and buried far more fibre-optic cable than anyone needed: by 2002 only about 2–5% of US fibre was in use. Global Crossing, which built a worldwide fibre network, went bankrupt on 28 Jan 2002, the biggest telecom failure up to then.

5

10 JAN 2000



AOL BUYS TIME WARNER

6

AOL had less than a fifth of Time Warner's revenue and staff, yet its shares were worth almost twice as much. So AOL used its sky-high shares as money to buy the media giant, in the biggest US merger at the time. In 2002 the merged company lost $98.7 billion, a corporate record, mostly by writing down what it had paid.

-$98.7BN LOSS, 2002



Everyone I know is getting rich. I'm buying too!

Prices were rising because prices were rising.



PRICED FOR PERFECTION

How expensive were shares? The Shiller CAPE compares share prices with 10 years of company profits, adjusted for inflation. Higher = hotter.



44.2 Dec 1999: NEW ALL-TIME RECORD

32.6 Sept 1929: the OLD record, just before the Great Crash

20.2 January 1995, the boom begins

14.8 Long-run average, 1881–1995



WHY 1929? It was the highest CAPE ever seen before the 1990s. Weeks later came the Wall Street Crash and the Great Depression. In 1999 that old danger mark was beaten by more than a third.

A CAPE of 44 means paying $44 for every $1 of yearly profit (averaged over 10 years). At the long-run average, people paid about $15.

Our "stocks expensive" light (CAPE above 25) had been ON for 43 months when the market peaked.

So everyone KNEW shares were expensive?



Yes! Seeing that shares are expensive is easy. Knowing WHEN prices will fall is the hard part.



TIMELINE PART 1: UP, UP...

Read from the bottom up. Green = good news, red = trouble, blue = a warning. Red stamp = the crash.

LATER

24 MAR 2000

S&P 500 PEAKS

1,527.46, the high for the wider US market.



20 MAR 2000

"BURNING UP"

Barron's: 51 internet firms could run out of cash within a year.



CRASH STARTS

10 MAR 2000

THE PEAK

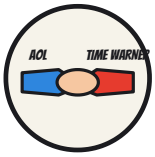
Nasdaq closes at 5,048.62. It won't see this level again for 15 years.



10 JAN 2000

AOL + TIME WARNER

Biggest US merger at the time, paid in AOL shares.



31 DEC 1999

NASDAQ +86% IN A YEAR

It ends 1999 at 4,069. Y2K passes quietly.



30 JUN 1999

FED STARTS RAISING RATES

First of six rises: 4.75% up to 6.5% by May 2000.



5 DEC 1996

"IRRATIONAL EXUBERANCE"

Greenspan's warning. Markets shrug it off.



8 FEB 1996

TELECOM ACT SIGNED

Phone and cable markets open to new competitors.



9 AUG 1995

NETSCAPE IPO

Shares double on day one. The internet gold rush begins.



EARLIER

PART 2: ...AND DOWN!

Read from the bottom up. The crash proper: 10 Mar 2000 to 9 Oct 2002 (red stamps).

LATER

9 OCT 2002 — CRASH ENDS

THE BOTTOM

Nasdaq 1,114.11 (-78%). S&P 500 776.76 (-49%).



21 JUL 2002

WORLDCOM BANKRUPT

Biggest US bankruptcy at the time, after billions in false profits.



2 DEC 2001

ENRON COLLAPSES

Fake accounts at a Wall Street favourite. Trust starts to crack.



11 SEP 2001

TERROR ATTACKS

Markets close for 4 trading days. Nasdaq -6.8% when they reopen.



MAR 2001

RECESSION BEGINS

The US economy shrinks until November 2001.



3 JAN 2001

FED STARTS CUTTING

Eleven cuts in 2001 take rates from 6.5% to 1.75%.



1 NOV 2000

PETS.COM SHUTS DOWN

268 days after its IPO. Its sock-puppet mascot becomes a symbol of the bust.



16 MAY 2000

RATES HIT 6.5%

The Fed's last rise. Borrowing is now expensive.



14 APR 2000 — THE CRASH

BLACK FRIDAY FOR TECH

Nasdaq -9.7% in one day. Now -34% from the peak in just 5 weeks.



3 APR 2000

MICROSOFT TALKS FAIL

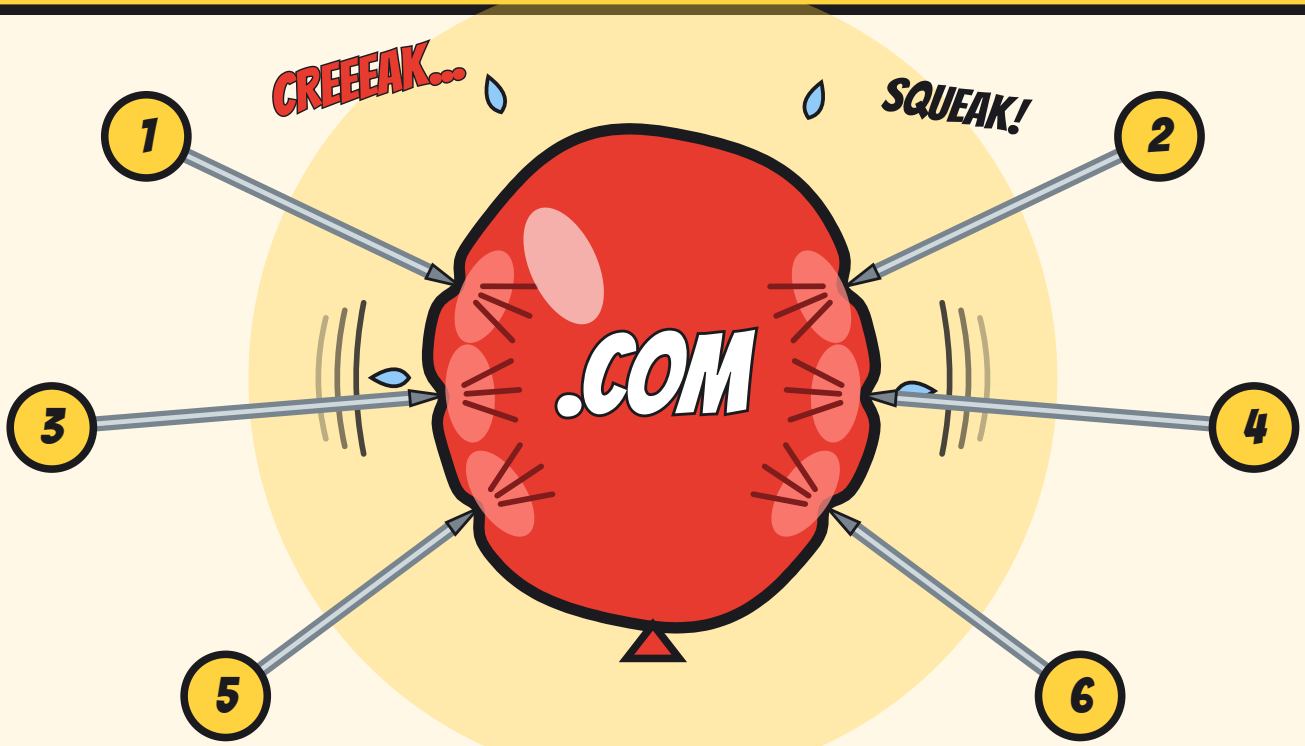
Settlement talks collapse; a judge rules it broke antitrust law. Nasdaq -7.6% that day.



EARLIER

WHAT POPPED IT?

No single event burst the bubble. Six pins pressed in, one after another:



1 DEARER MONEY

The Fed raised rates six times, from 4.75% to 6.5% (Jun 1999–May 2000). Future profits were suddenly worth less.

2 CASH RAN OUT

Most dot-coms lost money. When investors stopped funding them, they had months, not years, of cash left.

3 INSIDERS COULD SELL

Founders and staff are usually barred from selling for about 6 months after an IPO. After the record IPOs of 1999, those lock-ups ended in 2000.

4 REALITY CHECK

The Microsoft ruling and the Barron's cash study in spring 2000 made investors ask: who will actually make money?

5 SPENDING STOPPED

Firms that had over-built cut back hard. Business spending on computers and other IT equipment fell 24% (late 2000 to late 2002).

6 TRUST BROKE

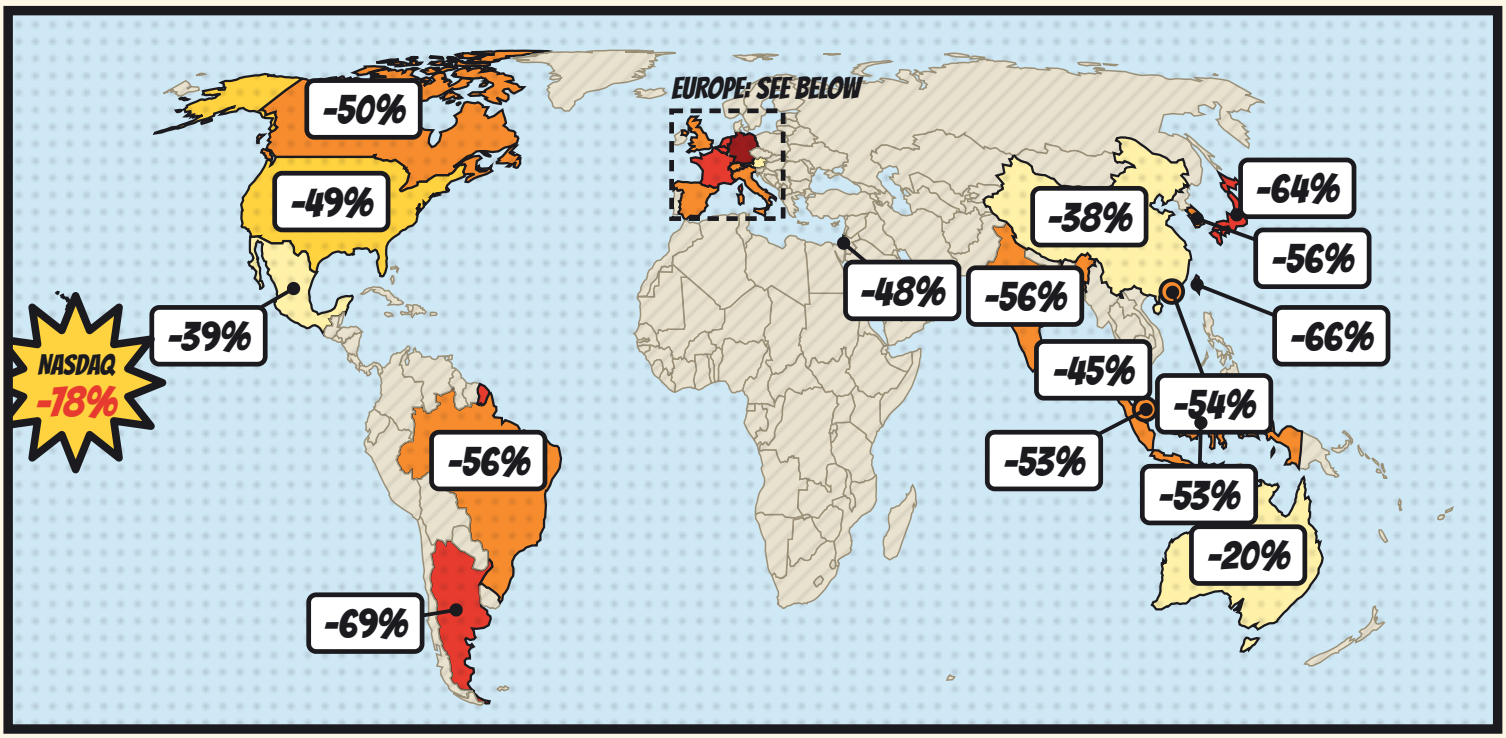
Enron (2001) and WorldCom (2002) showed some profits were fake. Investors began to doubt every company's accounts.

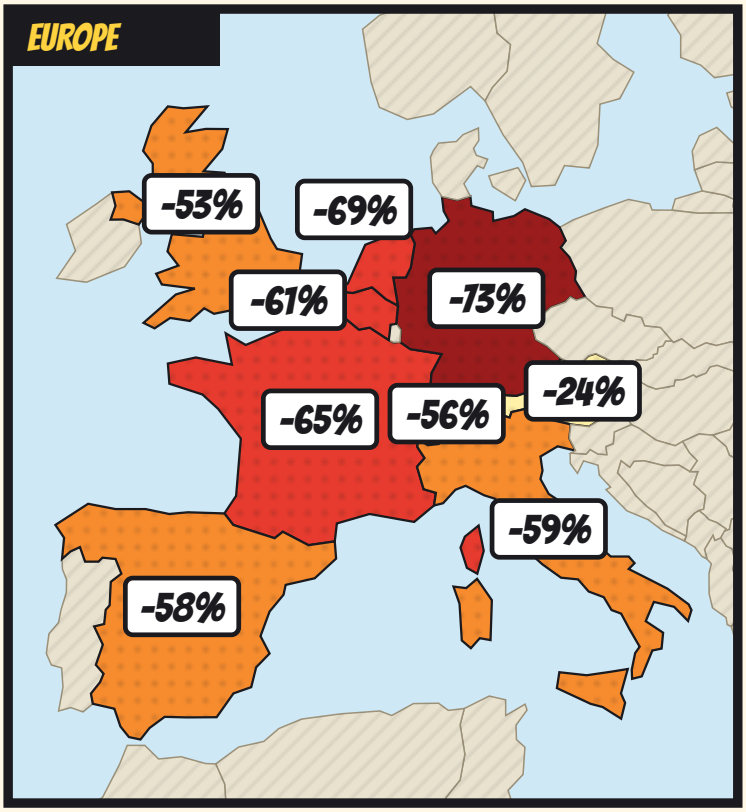


Our warning lights: credit spreads and company borrowing switched on 18 months before the peak. The Fed-hiking light only fired 3 months after it.

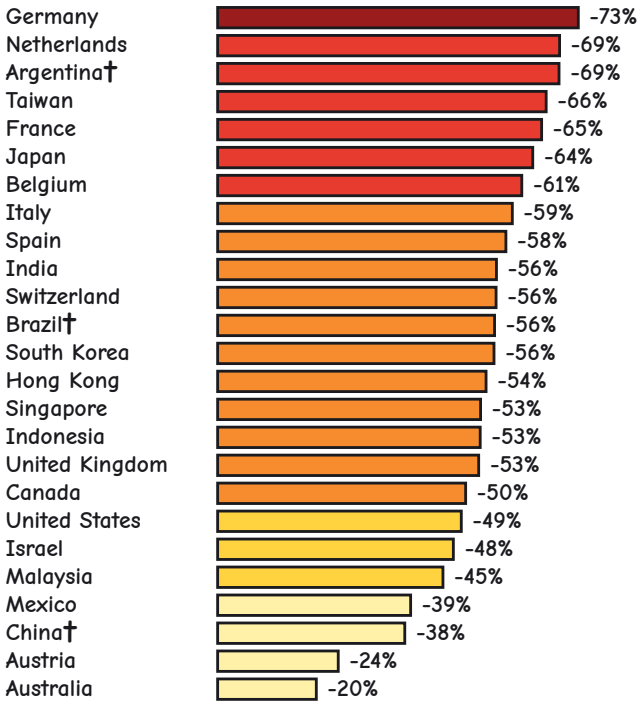
WHO GOT HURT?

When the bubble burst, share prices fell around the world. Each country is shaded by how far its main stock index dropped, from its 1999–2000 high to its lowest point by mid-2003.





THE FALLS, RANKED



INDEX FALL: fell 70% or more | 60–70% | 50–60% | 40–50% | less than 40% | no data used



Local-currency index closes, Yahoo Finance. Map: Natural Earth. † Argentina also had its own 2001 debt default; Brazil its own 2002 crisis; China = mainland market, then largely closed to foreign investors.

FIXING THE RULES

After the crash and the accounting scandals, governments rewrote the rules for companies, auditors and banks.



30 JUL 2002

UNITED STATES

SARBANES-OXLEY ACT

Bosses must personally sign off their company's accounts. A new watchdog, the PCAOB, checks the auditors.



28 APR 2003

UNITED STATES

GLOBAL RESEARCH SETTLEMENT

10 big Wall Street banks paid about $1.4 billion and had to separate stock research from deal-making.



28 JAN 2003

EUROPEAN UNION

MARKET ABUSE DIRECTIVE

One set of EU rules against insider dealing and market manipulation.



19 JUL 2002

EUROPEAN UNION

IFRS FOR LISTED FIRMS

From 2005, listed EU companies report with the same international accounting standards.



26 FEB 2002

GERMANY

CORPORATE GOVERNANCE CODE

Good-governance rules for listed firms: comply, or explain why not.



5 JUN 2003

GERMANY

NEUER MARKT CLOSED

Germany's dot-com market, whose index lost over 95% from its 2000 peak, was shut down.



20 JAN 2003

UNITED KINGDOM

HIGGS REVIEW

Led to a stricter Combined Code: at least half of a board should be independent directors.



30 MAR 2004

CANADA

CEO/CFO CERTIFICATION (52-109)

Canadian bosses must certify their filings, closely following the US rules.



2006 / 2008

JAPAN

"J-SOX" (FIEA 2006)

Listed companies must report on their internal controls, from fiscal years starting April 2008.



Notice something? Most of these rules target CHEATING: fake accounts and biased research. No law can stop people paying too much for exciting shares.

So spotting the next bubble is up to us?



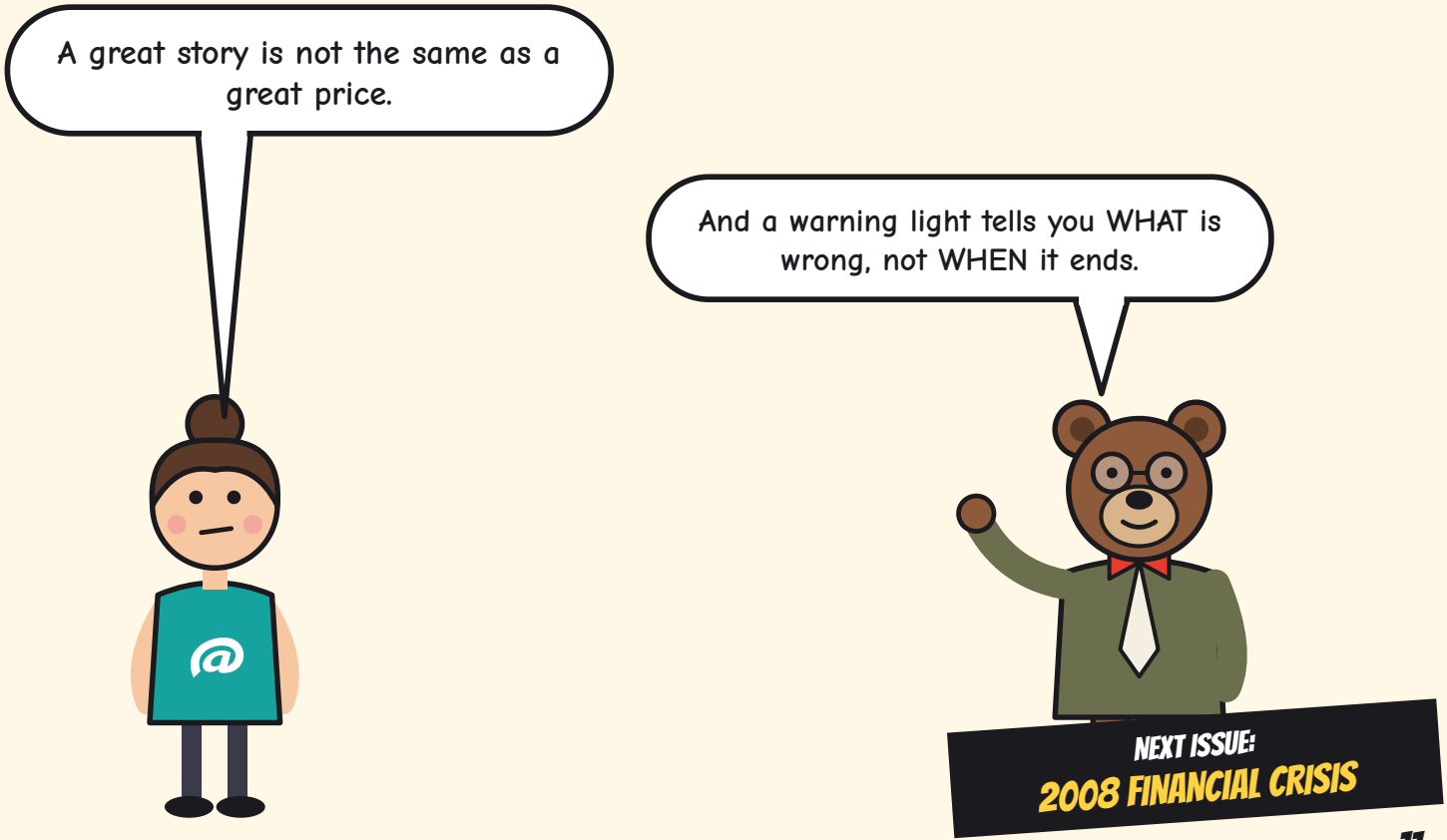
WHAT OUR WARNING LIGHTS SAID

The same 10 rules we use for every crash, applied to the months before the March 2000 peak.

YIELD CURVE INVERTED TOO LATE 5 months after	**FED HIKING FAST** TOO LATE 3 months after	**INFLATION HIGH** NO SIGNAL did not fire	**OIL SHOCK** WARNED 7 months early	**CREDIT SPREAD WIDENING** WARNED 18 months early
CORPORATE DEBT BUILD-UP WARNED 18 months early	**STOCKS EXPENSIVE** LONG ON on for 43 months	**PROFITS FALLING** TOO LATE 3 months after	**HOUSING SLUMP** NO SIGNAL did not fire	**FACTORY HOURS CUT** TOO LATE 5 months after

3 of 10 lights warned early: credit spreads and company borrowing 18 months ahead, oil 7 months ahead. Valuations had been flashing for years. The yield curve, Fed hikes and falling profits only showed up after the peak.

THE LESSON: A bubble can be obvious for years — the valuation light was on long before — but its timing is not. Rising corporate borrowing and bond spreads gave the nearer warning.



NEXT ISSUE:
2008 FINANCIAL CRISIS

SOURCES & NOTES

Every number in this comic was checked against the sources below on 21 October 2026. Index falls use the highest daily close in 1999–2000 and the lowest close after it up to June 2003, in local currency. Productivity, jobs, prices, the budget and interest rates come from official US data (BLS, BEA, Treasury, Federal Reserve) via FRED. Characters are fictional. For education only, not investment advice.

THE TECH REVOLUTION (PAGE 2)

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THE GOOD TIMES (PAGE 3)

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FROM BOOM TO BUBBLE (PAGE 4)

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FIXING THE RULES (PAGE 10)

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TIMELINE, TRIGGERS AND DATA

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https://fred.stlouisfed.org/series/OPHNFB (output per hour chart)
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https://www.nber.org/research/data/us-business-cycle-expansions-and-contractions
https://shillerdata.com/
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https://www.naturalearthdata.com/ (map outlines, public domain)